

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 13
Simulation of Distance Vector Routing Protocol

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Aim:

Simulation of **Distance Vector Routing** Protocol

Code:

```
set ns [new Simulator]
```

```
$ns color 1 red $ns color 2 blue
```

```
set n0 [$ns node] set n1 [$ns node] set n2 [$ns node] set n3 [$ns node] set n4  
[$ns node] set n5 [$ns node] set n6 [$ns node] set n7 [$ns node] set n8 [$ns  
node] set n9 [$ns node] set n10 [$ns node] set n11 [$ns node]
```

```
set nr [open out.tr w] $ns trace-all $nr  
set nf [open out.nam w] $ns namtrace-all $nf
```

```
proc finish { } { global ns nr nf $ns flush-trace close $nf
```

```
close $nr  
exec nam out.nam & exit 0  
}
```

```
$ns duplex-link $n0 $n1 $ns duplex-link $n1 $n2 $ns duplex-link $n2 $n3 $ns
```

duplex-link \$n3 \$n4 \$ns duplex-link \$n4 \$n5 \$ns duplex-link \$n5 \$n6 \$ns
duplex-link \$n6 \$n7 \$ns duplex-link \$n7 \$n8

\$ns duplex-link \$n0 \$n8

\$ns duplex-link \$n1 \$n10 1Mb 10ms DropTail \$ns duplex-link \$n0 \$n9 1Mb
10ms DropTail

1Mb 10ms DropTail 1Mb 10ms DropTail 1Mb 10ms DropTail 1Mb 10ms DropTail
1Mb 10ms DropTail 1Mb 10ms DropTail 1Mb 10ms DropTail 1Mb 10ms DropTail

1Mb 10ms DropTail

\$ns duplex-link \$n9 \$n11 1Mb 10ms DropTail \$ns duplex-link \$n10 \$n11 1Mb
10ms DropTail \$ns duplex-link \$n11 \$n5 1Mb 10ms DropTail

\$ns duplex-link-op \$n0 \$n1 orient right
\$ns duplex-link-op \$n1 \$n2 orient down
\$ns duplex-link-op \$n2 \$n3 orient down
\$ns duplex-link-op \$n3 \$n4 orient left
\$ns duplex-link-op \$n4 \$n5 orient left
\$ns duplex-link-op \$n5 \$n6 orient left
\$ns duplex-link-op \$n6 \$n7 orient left
\$ns duplex-link-op \$n7 \$n8 orient up
\$ns duplex-link-op \$n8 \$n0 orient up
\$ns duplex-link-op \$n5 \$n11 orient up

\$ns duplex-link-op \$n0 \$n9 orient right-down \$ns duplex-link-op \$n9 \$n11 orient right

\$ns duplex-link-op \$n11 \$n10 orient right \$ns duplex-link-op \$n1 \$n10 orient left-down

set udp0 [new Agent/UDP] \$ns attach-agent \$n0 \$udp0 \$udp0 set fid_ 1

set cbr0 [new Application/Traffic/CBR] \$cbr0 set packetSize_ 1000
\$cbr0 set rate_ 0.10mb
\$cbr0 set random_ true

\$cbr0 attach-agent \$udp0

set null0 [new Agent/Null] \$ns attach-agent \$n5 \$null0 \$ns connect \$udp0 \$null0

set udp1 [new Agent/UDP] \$ns attach-agent \$n1 \$udp1 \$udp1 set fid_ 2

set cbr1 [new Application/Traffic/CBR] \$cbr1 set packetSize_ 1000
\$cbr1 set rate_ 0.10mb
\$cbr1 set random_ true

\$cbr1 attach-agent \$udp1 set null1 [new Agent/Null]

\$ns attach-agent \$n5 \$null1 \$ns connect \$udp1 \$null1

\$ns rtpproto DV

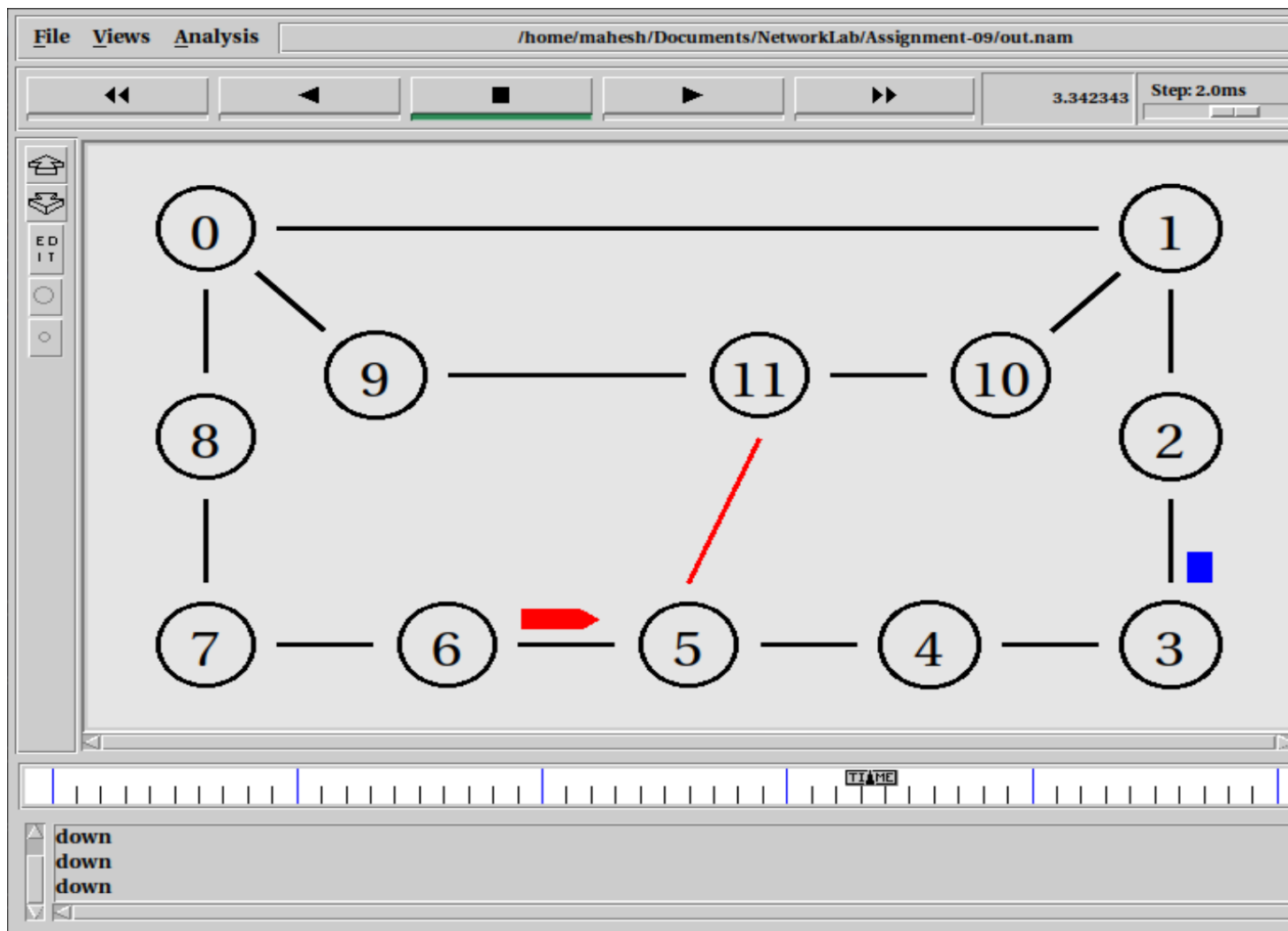
\$ns rtmodel-at 3.0 down \$n11 \$n5 \$ns rtmodel-at 2.0 down \$n7 \$n6 \$ns
rtmodel-at 4.0 up \$n11 \$n5 \$ns rtmodel-at 3.0 up \$n7 \$n6 \$ns at 0.0 "\$cbr0
start"

\$ns at 0.75 "\$cbr1 start"

\$ns at 5.0 "finish"

\$ns run

Output:



Learning Outcomes:

I learned how to implement Distance Vector Routing protocol.